Abdelaali Safir

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Education

École nationale supérieure d'électricité et mécanique (ENSEM) — Casablanca, Morocco
Engineering Degree in Computer Science — Grade: 14.7 — *Sept 2024 – expected July 2027*

Professional Experience

Pluriel — July – August 2026
Data Engineering Intern — *Marrakech, Morocco*

- Design of a **Change Data Capture** architecture with **Debezium**, **Kafka** and **ZooKeeper** to capture PMS database events in real-time, with operational **KPI** aggregation via **Faust**.
- Implementation of a generative AI pipeline with **Ollama**, **Langchain** and fine-tuned **Gemma 4** for **NLP-to-SQL** translation, integrated into a **FastAPI** API for contextual KPI queries.
- Development of a **Next.js** interface with real-time monitoring, full **Docker** containerization and orchestration of the entire pipeline for a hospitality decision-making chatbot.

Havet Digital Group (iaweb.dev) — June 2025
Data Science Intern — *Marrakech, Morocco*

- Design of a **Machine Learning** model benchmarking platform with **Gradio** to analyze and recommend ranking algorithms (**Wilson score**, K-Means, DBSCAN, GMM and hierarchical clustering) adapted to B2B contexts.
- Building a complete **ETL** pipeline: extraction of **NDJSON** data from web scraping, transformation to **CSV**, automated data preprocessing and statistical performance benchmarking.
- Development of an interactive testing interface with advanced visualizations allowing project managers to compare models.
- Bilingual presentation (French/English) of the software architecture and results — evaluation score: **17/20**.

Projects

E2E Micro-Batches Stock Market Data Pipeline

- Implementation of an end-to-end data pipeline from ingestion to visualization: real-time API ingestion via **Kafka**, micro-batch storage on **MinIO (S3-compatible)**, advanced orchestration with **Airflow** (Webserver, Scheduler, Metadata DB).
- Storage architecture on **Snowflake** (Medallion: Raw, Cleaned, Production) fed by modularized transformations with **dbt**, ensuring reliable schema deployment.
- Full deployment of services in a **Docker containerized environment**, integrating a monitoring layer (**Kafdrop**, **Zookeeper**), and interactive data visualization with **PowerBI**.

PFA Project: SIR + Deep Learning for Rumour Detection

- Processing and analysis of textual data with **Pandas**, **NumPy** and **Scikit-learn** (TF-IDF, vectorization), comprehensive ML model benchmarking and performance evaluation.
- Fine-tuning a **BERT** model (**covid-twitter-bert-v2**) with **PyTorch** and **Hugging Face** on **Google Colab (T4-GPU)** for tweet classification, surpassing classical models (**Logistic Regression, XGBoost, Random Forest**) with an AUC-ROC of 0.93.
- Innovative coupling of the classifier with an epidemiological **SIR** model to simulate the viral propagation of rumours, with dynamic calculation of transmission parameters based on Twitter metadata.
- Deployment of an interactive **Streamlit** application for real-time analysis (manual and CSV batch), featuring optimized caching; version control with **Git/GitHub**.
- Successful defense with a score of **18/20**; clear presentation of project architecture, technical choices and experimental results, capturing the jury's attention and demonstrating mastery of the entire end-to-end pipeline.

Certifications

Oracle Cloud Infrastructure 2025: Certified AI Foundations Associate
Oracle Cloud Infrastructure 2025: Certified APEX Developer (currently enrolling this certificate)

July 25, 2026